Achouri Youcef Islem

Machine Learning Research Internship

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Final-year engineering student interested in speech, NLP and model implementation. Available January 2027.

EDUCATION

ESI-SBA 2021 - Present

École Nationale Supérieure d'Informatique de Sidi Bel Abbès
State Engineering Degree in Computer Science - fifth year, degree in progress.
Specialty: Artificial Intelligence and Data Structures (AIDS).

SELECTED PROJECTS

StethoSpeech Research Reproduction | Python, PyTorch

- Trained a 19.7M-parameter Transformer on paired non-audible murmur and speech features, reproducing a silent-speech research approach.
- Selected epoch 35 from 45 epochs; evaluated speech-feature prediction on 1,133 held-out utterances from familiar speakers (MAE 0.23171). Speech intelligibility remains unevaluated.

English to French Translation | PyTorch 2026

- Implemented Transformer attention, causal masking and positional encodings, with a custom byte-pair tokenizer and training pipeline.

Neural Network from Scratch | Python, NumPy 2026

- Implemented a two-layer MNIST classifier with vectorized forward propagation, analytical backpropagation and mini-batch gradient descent, without automatic differentiation.

Arabic Structured Information Extraction | Python, Qwen 2.5, Ollama 2025

- Authored the NLP module for the team's PlanMilt DZ academic GIS project, converting Arabic descriptions into structured scene data.
- Defined the extraction schema and prompts, with JSON parsing, validation and synthetic Arabic examples for local evaluation.

TECHNICAL SKILLS

Research programming: Python, PyTorch, NumPy, model training and evaluation.
ML foundations: Analytical backpropagation, gradient descent, Transformer attention.
Language and speech: Sequence modelling, BPE tokenization, speech-feature regression.
Engineering: Git, pytest, Linux, Docker, FastAPI, JSON Schema.

LEADERSHIP AND COMMUNICATION

Vice President | INGENIUMS Club, ESI-SBA 2023 - 2024

- Co-organized student technology events and programming contests, including Ingeneer, Ingehack and Ingecode.
- Spoke at ESI-SBA Welcome Day and demonstrated technical projects at campus exhibitions.

AWARDS

2nd place, DevFest Hackathon 2023 - GDG Sidi Bel Abbès, Team Infinity.
3rd place, NCS Hackathon 2024 - Numidia Computer Society / NiT.

Languages: English and French (strong proficiency).